

SWE 4304: SOFTWARE PROJECT LAB 1

TEAM 4

PROGRESS PRESENTATION

FEATHER DB





MEMBERS

WASI OMAR SYED

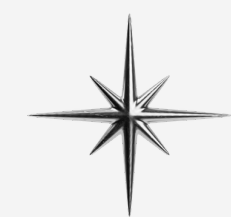
230042143

MD SAIF-AL SARKER

230042106

M SAFWAN HASAN KHAN

230042117



BRIEF OVERVIEW

CUSTOM SQL DATABASE ENGINE

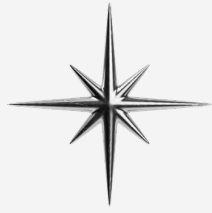
Lightweight SQL implementation
built from scratch

FILE BASED ACCESS CONTROL

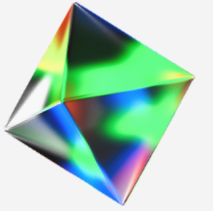
Data is stored on disk using file-
based storage for persistence

3 FUNCTIONAL COMPONENTS

Storage Manager, Query
Processor, Transaction
Manager



FEATURES



SQL PARSING & AST

Tokenizes SQL and builds
Abstract Syntax Tree

FILTERING & SORTING

WHERE clause filtering
and ORDER BY sorting

CRUD OPERATIONS

CREATE, READ,
UPDATE, DELETE
operations

INTERACTIVE REPL

Real-time SQL command
execution and feedback

NESTED QUERIES

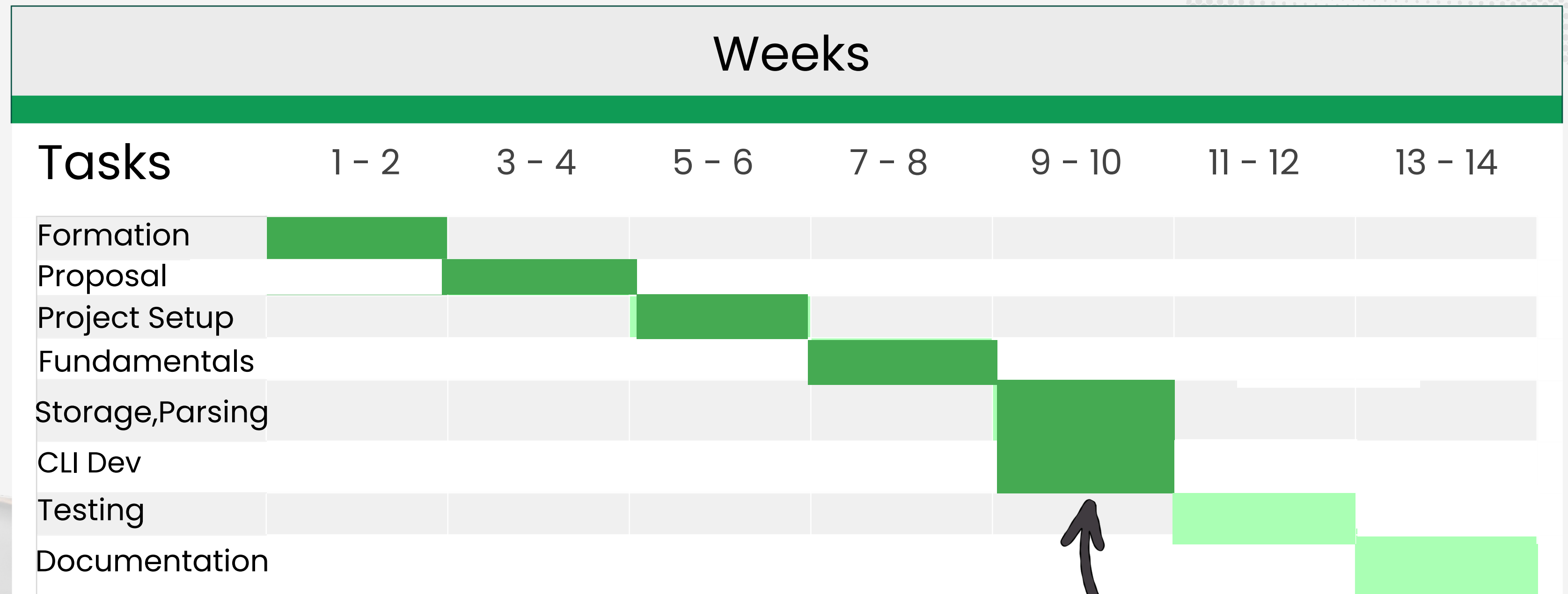
Supports subqueries in
SELECT, FROM, WHERE

PERSISTENT STORAGE

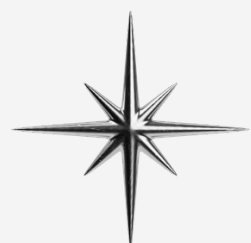
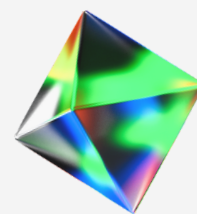
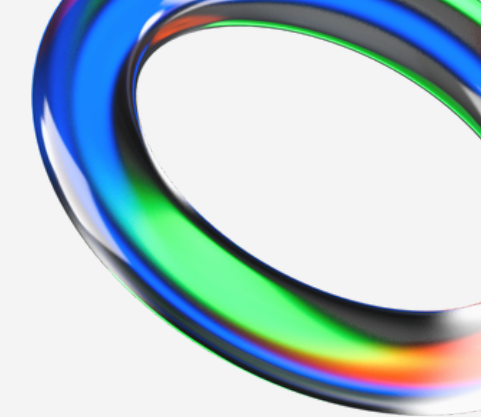
Data survives application
shutdown



TIME LINE



DEMONSTRATION



THANK YOU!

